

torch.as_strided#

https://pytorch.org/docs/stable/generated/torch.as_strided.html#torch.as_strided

Description:

Create a view of an existing `torch.Tensor` input with specified size, stride and `storage_offset`. Prefer using other view functions, like `torch.Tensor.view()` or `torch.Tensor.expand()`, to setting a view's strides manually with `as_strided`, as this function will throw an error on non-standard Pytorch backends (that do not have a concept of stride) and the result will depend on the current layout in memory. The constructed view must only refer to elements within the Tensor's storage or a runtime error will be thrown. If the generated view is "overlapped" (with multiple indices referring to the same element in memory), the behavior of inplace operations on this view is undefined (and might not throw runtime errors).

input (Tensor) – the input tensor. size (tuple or ints) – the shape of the output tensor

Function Signature

```
torch.as_strided(input, size, stride, storage_offset=None) →  
Tensor#
```

Parameters

Code Examples

```
>>> x = torch.randn(3, 3)
>>> x
tensor([[ 0.9039,  0.6291,  1.0795],
        [ 0.1586,  2.1939, -0.4900],
        [-0.1909, -0.7503,  1.9355]])
>>> t = torch.as_strided(x, (2, 2), (1, 2))
>>> t
tensor([[0.9039, 1.0795],
        [0.6291, 0.1586]])
>>> t = torch.as_strided(x, (2, 2), (1, 2), 1)
tensor([[0.6291, 0.1586],
        [1.0795, 2.1939]])
```

Notes & Warnings

WARNING

Warning Prefer using other view functions, like `torch.Tensor.view()` or `torch.Tensor.expand()`, to setting a view's strides manually with `as_strided`, as this function will throw an error on non-standard Pytorch backends (that do not have a concept of stride) and the result will depend on the current layout in memory. The constructed view must only refer to elements within the Tensor's storage or a runtime error will be thrown. If the generated view is "overlapped" (with multiple indices referring to the same element in memory), the behavior of inplace operations on this view is undefined (and might not throw runtime errors).

Detailed Documentation

Documentation

Create a view of an existing `torch.Tensor` input with specified size, stride and `storage_offset`.

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`input (Tensor)` – the input tensor.

`size (tuple or ints)` – the shape of the output tensor

`stride (tuple or ints)` – the stride of the output tensor

`storage_offset (int, optional)` – the offset in the underlying storage of the output tensor. If `None`, the `storage_offset` of the output tensor will match the input tensor.